

Overview of Cloud Platforms for GenAI



Content

- Introduction to Amazon Bedrock
- Why do we need Amazon Bedrock
- Key Benefits of Amazon Bedrock
- Introduction to Microsoft Azure OpenAI
- Key Benefits of Microsoft Azure OpenAI

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Introduction to Amazon Bedrock

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What is Amazon Bedrock?

Amazon Bedrock is a fully managed **Generative AI (GenAI)** service provided by AWS

- “Managed” means AWS takes care of infrastructure, scaling, and maintenance.
- It allows you to **use powerful pre-trained AI models directly through APIs**
- You do **not** need to train models from scratch.
- You only send input (prompt) and receive output (response).



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What is Amazon Bedrock?

Using Bedrock, you can build:

- **Chatbots** (customer support, assistants)
- **Text generators** (summaries, reports, content)
- **Image generators** (AI-created images)
- **AI agents** that can reason and take actions

You do not need to manage:

- Servers
- GPUs
- Model hosting
- Scaling for high traffic

“Amazon Bedrock = Ready-to-use AI models on AWS”

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Why Do We Need Amazon Bedrock?

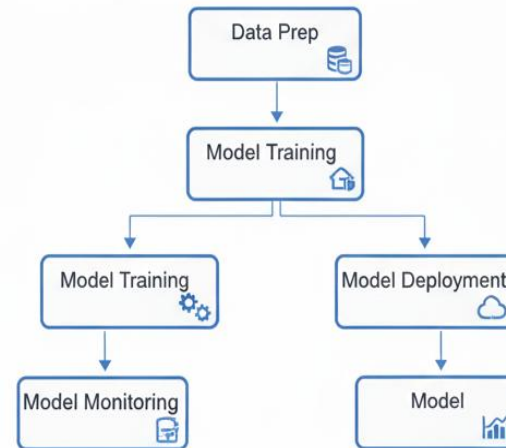
Building AI from scratch is:

- Expensive ☐
- Complex ☐ ☐
- Time-consuming ☐ ☐

Bedrock removes these challenges by:

- Providing **pre-trained foundation models**
- Offering **serverless access**
- Handling scaling automatically

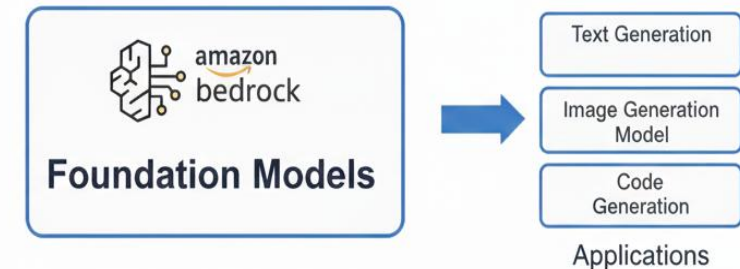
✗ Traditional ML Setup



Web:

Manual Infrastructure
Complex Orcherstration
High Overhead

✓ Amazon Bedrock Managed Approach



Simplified Development
Scalable & Serverless
Reduced Costs

What Are Foundation Models (FMs)?

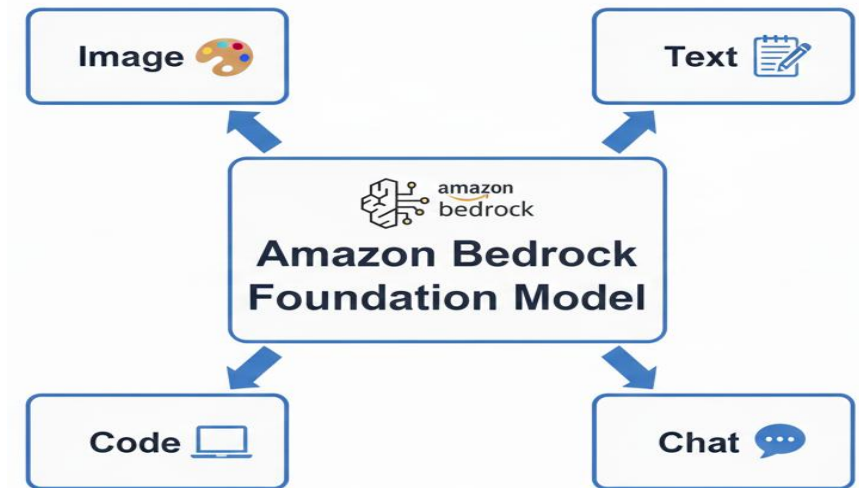
Foundation Models are large pre-trained AI models

- Trained on **huge amounts of data**
- Can perform many tasks with just prompts
- The same model can be reused for multiple applications
This makes development **faster, cheaper, and easier**.

Common Capabilities:

- **Text generation** – write articles, summaries, emails
- **Chat & Q&A** – conversational assistants and chatbots
- **Image generation** – create images from text descriptions
- **Code assistance** – generate, explain, or debug code

“One powerful model, many different tasks”



Foundation Models Available in AWS Bedrock

- **Amazon Titan** – Text, embeddings
- **Anthropic Claude** – Conversational AI
- **Meta Llama** – Open-source LLMs
- **Stability AI** – Image generation
- **AI21 Labs** – Text generation

What is Multi-Model Access in Bedrock?

Multi-model access means using multiple AI models from a single platform

- Amazon Bedrock gives access to models from different AI providers in one place.
- Instead of relying on one AI model for everything,
- You can **select the best model for each specific task.**

Different models are good at different jobs:

- Some are better at conversations
- Some are better at understanding text meaning
- Some specialize in image generation

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What is Multi-Model Access in Bedrock?

Example: One Application, Multiple Models

- **Claude** → Used for chatbots and natural conversations
- **Amazon Titan** → Used for text embeddings, search, and recommendations
- **Stability AI** → Used for generating images from text

All these models are accessed using the **same Amazon Bedrock service and APIs**
No need to integrate separate tools or platforms

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“Choose the right brain for the right job”

How Multi-Model Access Works (Architecture)

Why Multi-Model Switching is Easy

To switch models:

- You only change the **model ID** in the API call

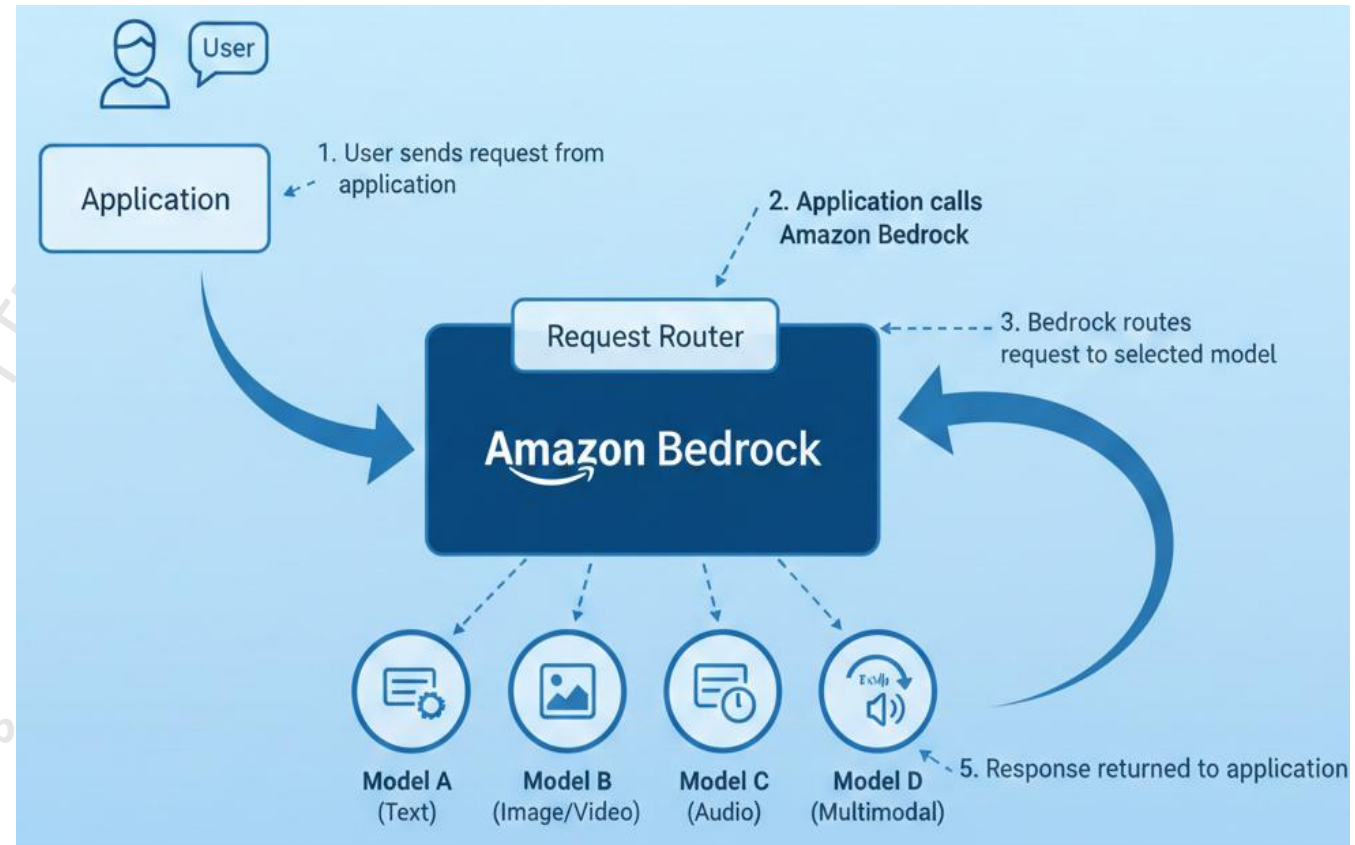
No need to:

- Rewrite application logic
- Change infrastructure
- Deploy new servers

Same code structure, different AI brain

No Infrastructure Management Required

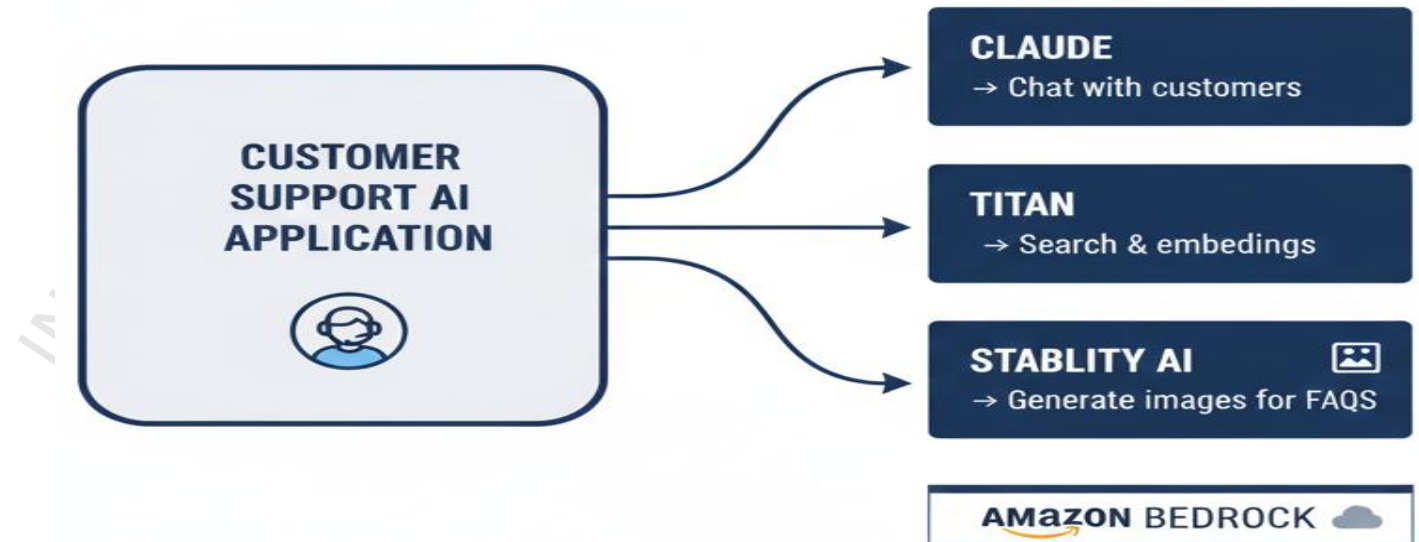
- AWS automatically handles:
 - Model hosting
 - Scaling with traffic
 - Load balancing
 - GPU management
 - High availability



"App talks to Bedrock, Bedrock talks to the model"

Example Use Case – Single App, Multiple Models

A company builds **one AI-powered customer support application** using Amazon Bedrock. Instead of using a single AI model for everything, the application uses **different models for different tasks**.



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BENEFITS:

- Better performance
- Cost optimization
- Flexibility

Example Use Case – Single App, Multiple Models

How Different Models Are Used :

1. Claude → Customer Chat & Q&A

- Handles natural conversation with users
- Understands customer questions
- Generates human-like responses
- Used for live chat and support tickets

2. Amazon Titan → Search & Embeddings

- Converts FAQs, manuals, and documents into embeddings
- Helps in **semantic search** (finding correct answers, not just keyword matches)
- Improves accuracy of responses given by the chatbot

3. Stability AI → Image Generation

- Generates images for:
 - Product setup guides
 - Troubleshooting steps
 - Visual FAQs
- Helps customers understand solutions faster

Key Benefits of Using Amazon Bedrock



SECURE

Data stays within AWS

Your data is not exposed to public systems



SCALABLE

For enterprise use

Same service works for startups and large enterprises



FULLY MANAGED & SERVERLESS

Easy integration with AWS services
Pay as you go pricing

You use the model, AWS manages everything behind the scenes.



FASTER INNOVATION

Rapid prototyping with leading FMS

You focus on ideas, Bedrock gives you the best AI brains instantly.



FLEXIBLE

Choice of leading models

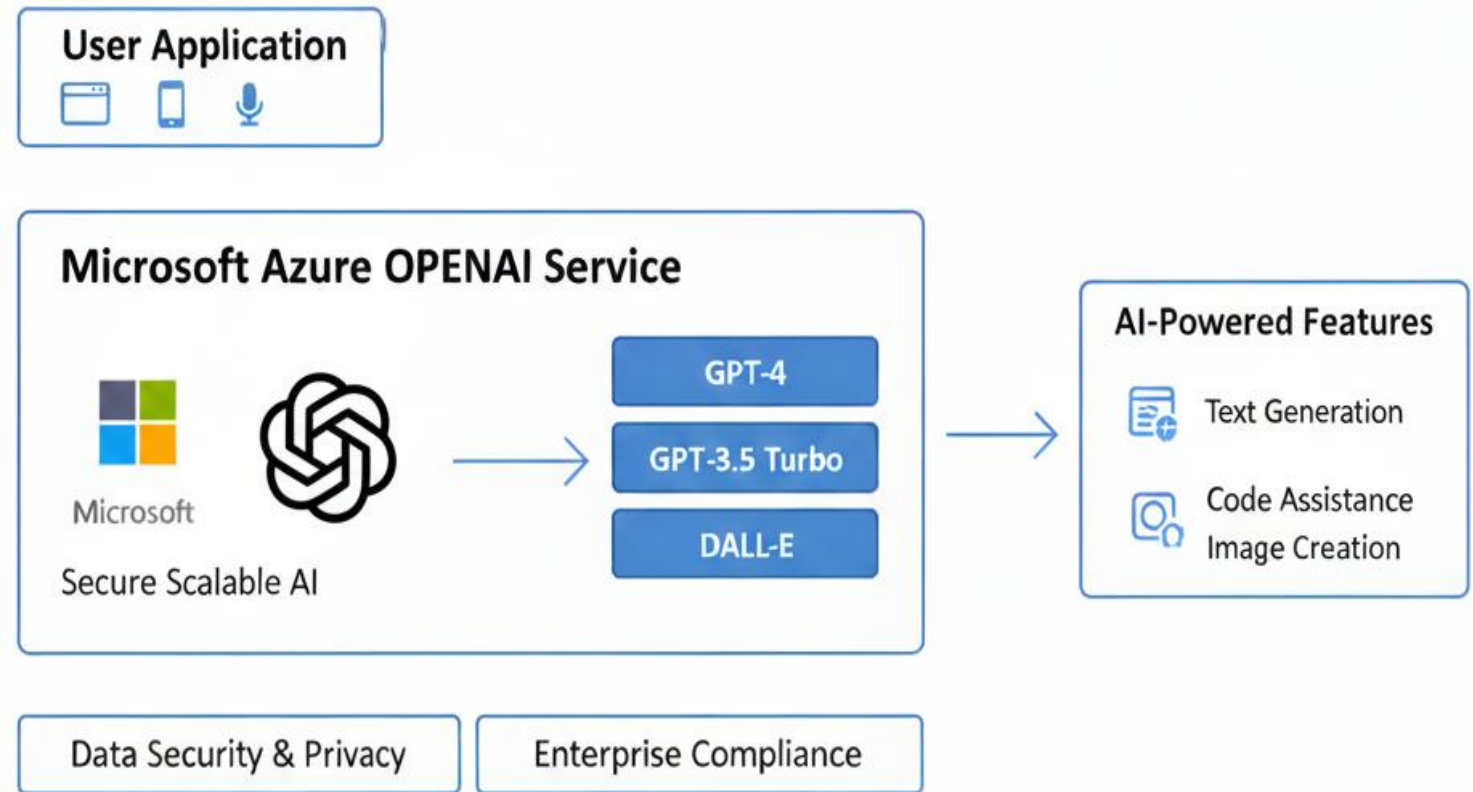
Amazon Bedrock provides easy access to the latest versions of leading foundation models

Introduction to Microsoft Azure OpenAI

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What is Microsoft Azure OpenAI?

- **Microsoft Azure OpenAI** is a **cloud service on Azure** that provides access to **OpenAI's powerful models**
- It allows organizations to use models like **GPT-4** securely on Azure
- Designed for **enterprise, security, and compliance**
- Used to build **chatbots, copilots, content generators, and AI assistants**



Azure OpenAI = OpenAI models hosted securely on Azure

Why Azure OpenAI? (Need & Purpose)

Direct use of public OpenAI APIs may not meet:

- Enterprise security needs
- Data privacy requirements

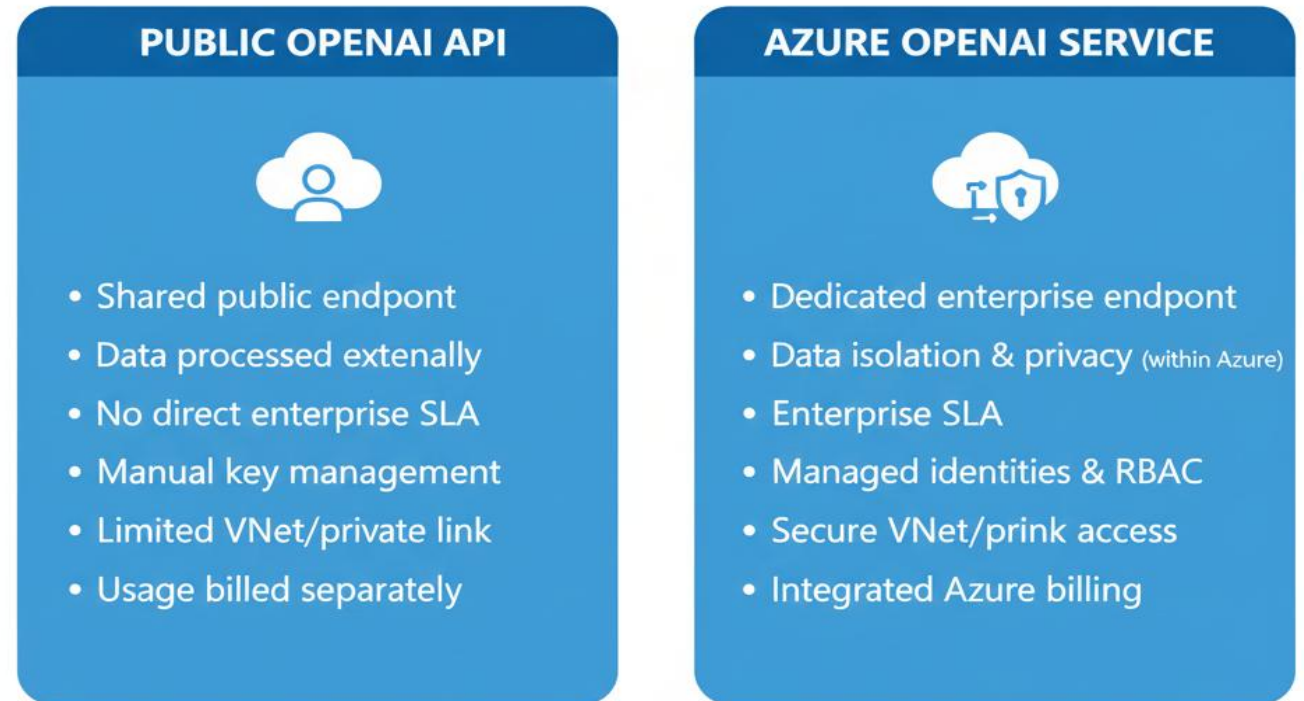
Azure OpenAI solves this by offering:

- Private deployments
- Enterprise-grade security
- Azure integration

Key Benefits:

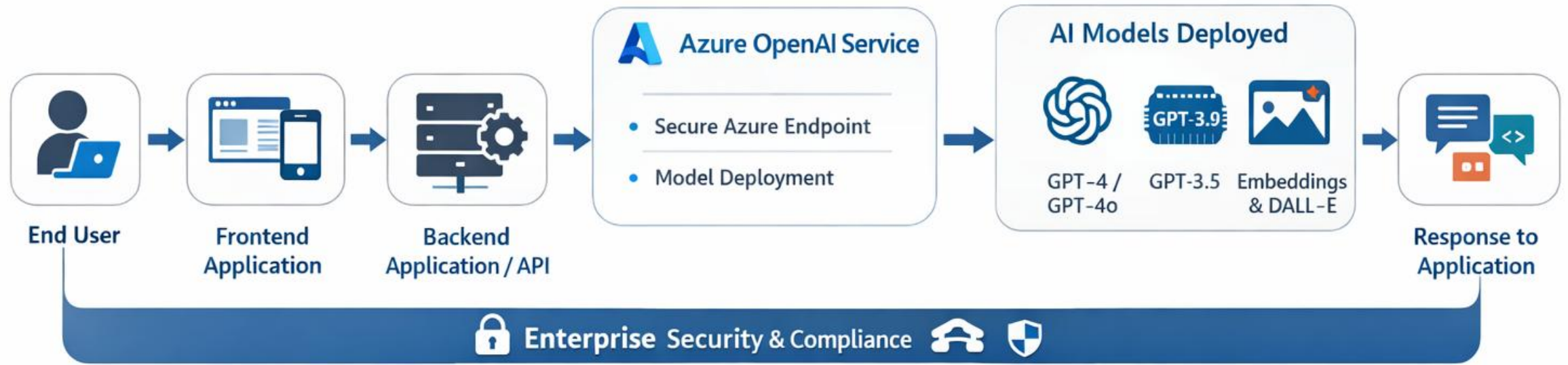
- Data stays within **Azure environment**
- No training on customer data by default
- Scalable and reliable

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FOR ENTERPRISE & PRODUCTION USE

How Azure OpenAI Works (High-Level Architecture)

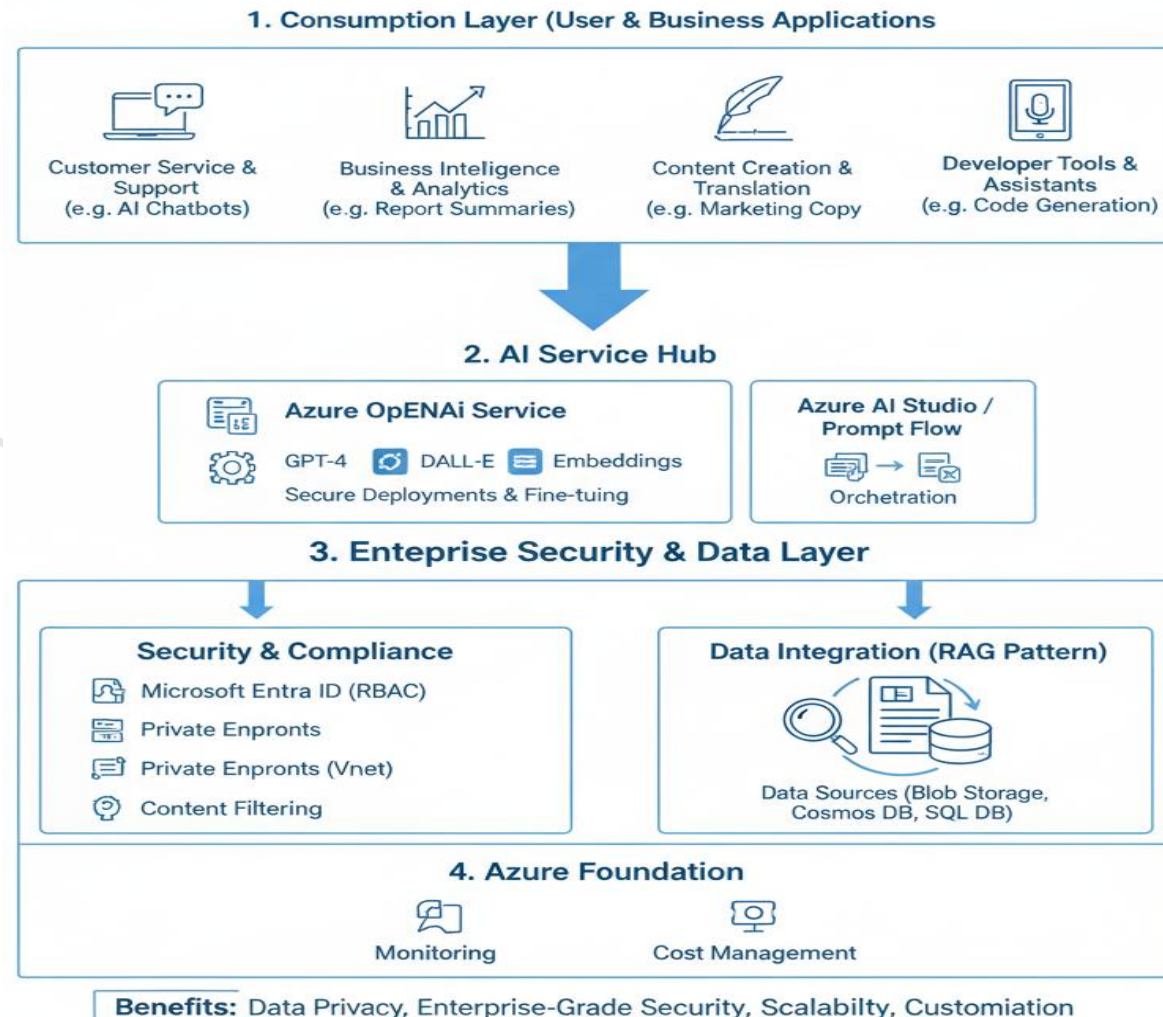


- User interacts with application
- Application sends request to Azure OpenAI endpoint
- Azure OpenAI routes request to deployed model (GPT-4, GPT-3.5, etc.)
- Model processes the prompt and returns the response

Advanced Use Cases with Azure OpenAI

- **Enterprise AI** across chat, documents, search, and code
- From **chatbots** to **enterprise copilots**
- **Intelligent assistants**, **smarter search**, automated knowledge
- Multiple AI use cases on **one secure platform**
- Turning **enterprise data** into **intelligent applications**

Website:



Security & Enterprise Features of Azure OpenAI

Data is not used to retrain models:

- Prompts, responses, and enterprise data remain private.
- Customer data is not shared or reused for training OpenAI models.

Built-in integration with enterprise security services:

- **Azure Active Directory (Entra ID)** for secure user authentication
- **Private networking** using VNET and private endpoints
- **Role-based access control (RBAC)** to restrict who can access or manage models

Designed for enterprise compliance:

- Meets common enterprise security and compliance standards.
- Suitable for regulated industries like finance, healthcare, and government.



Azure OpenAI = Secure GenAI built for enterprise environments

Key Benefits of Azure OpenAI



Access to Advanced OpenAI Models

- GPT-4, GPT-5, Embeddings & Image Models
- Chatbots, Copilots & Enterprise AI Assistants



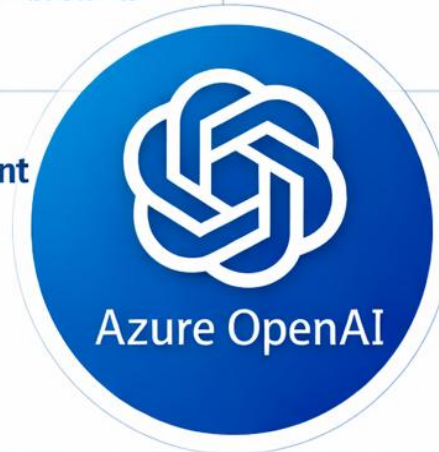
Enterprise-Grade Security & Privacy

- Data not used to retrain models
- AAD & RBAC Integration
- Private Networking & Compliance



Controlled Model Deployment

- Named Endpoints
- Usage & Cost Control
- Easy Versioning



Seamless Azure Integration

- Azure App Services
- Azure Functions
- Azure AI Search
- Azure Storage

Scalable & Reliable

- Auto Scaling & High Availability
- Managed Infrastructure
- Production Ready

AWS Bedrock vs Azure OpenAI – Key Differences

Feature / Aspect	AWS Bedrock	Azure OpenAI
Service Purpose	Managed GenAI service providing access to multiple foundation models	Enterprise cloud service to use OpenAI models securely on Azure
Model Access	Multi-model access via API	Model deployment + endpoint based access
Model Setup	No deployment required	Model must be deployed before use
Model Selection	Selected directly in API call (model ID)	Selected via deployment name
Supported Models	Amazon Titan, Anthropic Claude, Meta Llama, Stability AI, AI21	GPT-4, GPT-5, GPT-3.5, DALL·E, Embeddings
Multi-Model Usage	Very easy (switch models instantly)	Possible, but requires multiple deployments

AWS Bedrock = Plug-and-play access to multiple AI models

Azure OpenAI = Secure, enterprise-controlled deployment of GPT models

AWS Bedrock vs Azure OpenAI – Key Differences

Feature / Aspect	AWS Bedrock	Azure OpenAI
Scalability	Automatic scaling handled by AWS	Controlled scaling per deployment
Cost Control	Pay-per-use, less granular control	Fine-grained cost control per deployment
Versioning	Model version selected in API	Strong versioning via deployments
Security Model	AWS IAM, VPC integration	Azure AD, RBAC, private networking
Data Usage for Training	Customer data not used to train AWS models	Customer data not used to train OpenAI models
Best Use Case	Rapid prototyping, experimentation, multi-model comparison	Enterprise production apps, compliance-heavy workloads
Learning Curve	Very easy for beginners	Moderate (enterprise concepts involved)
Ideal Audience	Students, developers, startups	Enterprises, regulated industries

Thank You

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